MILITARY SPECIFICATION

PROJECTILE, 20-MILLIMETER, MULTIPURPOSE TRACE SELF-DESTRUCT, M940 METAL PARTS

This specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and preparation for delivery of metal parts for Projectile 20mm, Multipurpose, Trace Self-Destruct M940.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-A-70625 - Automated Acceptance Inspection Equipment for Non-Electronic Components, Design, Testing and Approval, of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-651 - Visual Inspection Standards for 20MM Ammunition and Components
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1190- Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (SEE 6.7)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

7258928 - Marking on M548 Shipping and Storage Container for 20mm Cartridges
12551767 - Cartridge, 20mm, MPT-SD, M940
12551771 - Disc, Closure Nozzle
12551773 - Nose Cap
12551778 - Seal
12551779 - Disc, Metering
12551780 - Projectile Body Assembly, Painted
12551781 - Projectile Body Assembly
12551782 - Body, Projectile
12551783 - Band, Rotating

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INSPECTION EQUIPMENT DRAWINGS

11075203	-	Eddy Current Flaw Detector
12551784	-	Non-Destructive Test Standard

PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505-	Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges.

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI Z1.1	-	Guide for Quality Control Charts
ANSI Z1.2	-	Control Chart Method of Analyzing Data
ANSI Z1.3	-	Control Chart Method of Controlling Quality During Production

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117	-	Standard Method of Salt Spray (fog) Testing
ASTM-E18	-	Method of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 General.

3.1.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.1.2 Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without prior approval (6.4).

3.2 Components. All parts and assemblies shall comply with all requirements specified on Drawing (Dwg.) 12551780, all associated drawings, and with all requirements specified in applicable specifications.

3.3 First article. When specified in the contract or purchase order, a sample shall be subjected to a first article inspection in accordance with the technical provisions herein (see 4.3 and 6.5).

3.4 Banding. The rotating band blank and the rotating band seat shall be clean at the time of banding. Shot or sandblasting shall not be used to clean the rotating band seat after knurling. The rotating band blank shall fill the band seat to include a minimum 50% of the volume of the dovetail undercut.

3.5 Soundness. Metal discontinuity in the projectile body as determined by eddy current inspection shall be less than that represented by non-destructive test standard (dwg. 12551784).

3.6 Protective coating.

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3.6.1 Phosphate coating. The pretreatment cleaning process shall be in accordance with the applicable requirements of dwg. 12551780. The phosphate coating shall be in accordance with the requirements of TT-C-490 for appearance and minimum weight as applicable to the type of phosphate coating used. Test specimens (panels) shall be in accordance with the requirements of TT-C-490.

3.6.2 Paint coating. The exterior coating of the projectile shall be in accordance with the applicable requirements of dwg. 12551780. The exterior paint adhesion shall be in accordance with applicable requirements of TT-C-490.

3.6.3 Corrosion resistance. The exterior painted surface of the projectile shall show no evidence of corrosion (red rust) after exposure to a 5 percent solution of atomized salt spray for 24 hours.

3.7 Metal parts security. The projectile metal parts when assembled into test cartridges in accordance with dwg. 12551767 (inert loaded projectile) shall not separate or break up during testing for metal parts security (see 6.8).

3.8 Workmanship. The requirements for workmanship are as specified by applicable drawings, referenced specifications and the following:

3.8.1 Processing defects. The Projectile shall be free of defects, such as dents, deep scratches, cuts, excessive surface roughness, metal flaws, burrs, sharp edges and other forms of substandard workmanship.

3.8.2 Cleanliness. The projectile shall be free of chips, oil and all other detrimental contaminants.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of any defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection (4.3)

b. Quality Conformance Inspection (4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

a. Inspection and nonfiring tests:

Part Description	Drawing	Quantity	Method Para
Standard test panels		3	4.5.3
Band, rotating	12551783	32	4.4.2.1
Body, Projectile (unbanded)	12551782	125	4.4.2.2
Body, Projectile assembly (banded body, uncoated)	12551781	125	4.4.2.3
Projectile, body assembly, painted (banded body, painted)	12551780	125	4.4.2.4
Disc, closure nozzle	12551771	125	4.4.2.5
Nose, cap	12551773	125	4.4.2.6
Seal	12551778	125	4.4.2.7
Disc, Metering	12551779	125	4.4.2.8

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b. Firing tests:

Part Description	Drawing	Quantity	Method Para
Projectile, Body assembly, painted	12551780	4000	4.5.1

c. The projectile, body assembly, painted, dwg. 12551780 shall be identified in accordance with MIL-STD-1168.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078. Failure of the original first article sample to be in accordance with the applicable drawings and requirements of Table I, as specified herein will result in sample disapproval.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70794(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Projectile, 20mm, MPT-SD, M940				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Band, Rotating (Dwg. 12551783)				
	Examination for defects		ACC-0		
		32	REJ-1	3.2	4.4.2.1
	Body, Projectile (Dwg 12551782)				
	Examination for defects		ACC-0		
		125	REJ-1	3.2	4.4.2.2
	Hardness	10	ACC-0		
			REJ-1	3.2	4.5.6.2
	Projectile Body Assembly (Dwg. 12551781)				
	Examination for defects	125	ACC-0		
			REJ-1	3.2	4.4.2.3
	Band fill	20	ACC-0		
			REJ-1	3.4	4.5.2
	Painted, Projectile Body Assembly (Dwg. 12551780)				
	Examination for defects	125	ACC-0		
			REJ-1	3.2	4.4.2.4
	Phosphate coating	3	ACC-0		
			REJ-1	3.6.1	4.5.3
	Paint adhesion	13	ACC-0		
			REJ-1	3.6.2	4.5.4
	Corrosion resistance	32	ACC-0		
			REJ-1	3.6.3	4.5.5

NOTES:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-P-70794(AR)

PARAGRAPH	TITLE: Projectile, 20mm, MPT-SD, M940		SHEET 2 OF 2		DRAWING NUMBER: SEE BELOW; NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Disc, Closure Nozzle (Dwg. 12551771)				
	Examination for defects	125	ACC-0 REJ-1	3.2	4.4.2.5
	Nose Cap (Dwg 12551773)				
	Examination for defects	125	ACC-0 REJ-1	3.2	4.4.2.6
	Hardness	80	ACC-0 REJ-1	3.2	4.5.6.1
	Seal (Dwg. 12551788)				
	Examination for defects	125	ACC-0 REJ-1	3.2	4.4.2.7
	Disc, Metering (Dwg. 12551779)				
	Examination for defects	125	ACC-0 REJ-1	3.2	4.4.2.8
	Firing Test				
	Metal parts security	4000	ACC-0 REJ-1	3.7	4.5.1

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality Conformance Inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Major and minor defects. To determine product acceptability, major and minor defects as listed herein may be considered collectively on a class basis or defects may be considered individually. When a class basis is utilized the Acceptable Quality Level (AQL) for major defects shall be 1.5 percent and the AQL for minor defects shall be 2.5 percent. However, when major defects are considered collectively on a class basis, the AQL for any individual defect shall be limited to .40 percent. In addition, where three or less defects appear in a classification sampling plans applicable to a class shall not be used for major defects (see 6.6).

c. Special defect. Unless otherwise specified, one hundred percent examination shall be performed for all special defects.

d. Visual defects. MIL-STD-651 shall apply in defining and evaluating visual defects.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70794(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551783
4.4.2.1	Band Rotating				NEXT HIGHER ASSEMBLY 12551781
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Foreign matter		.25	3.8.2	Visual
102	Defective metal, including lamination		.25	3.2	Visual
103	Incomplete or deformed		.25	3.2	Visual 1/
Minor					
201	.095 Width		.40	3.2	Gage
202	Poor workmanship		.65	3.8	Visual

NOTES: 1/ A rotating band blank which is deformed or is judged to lack sufficient metal in any part of its width or thickness to provide for complete band seat fill and minimum rotating band profile dimensions, shall be classed defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70794(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 12551782
4.4.2.2	Body, Projectile				NEXT HIGHER ASSEMBLY 12551781
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Knurling missing or incomplete		.25	3.2	Visual
102	Foreign material in band seat		.25	3.2/ 3.8.2	Visual
103	.012 min, Depth of knurl		.25	3.2	Visual 1/
104	1.36 Length, mouth to rear of band seat		.25	3.2	Gage
105	.784 Diameter bourrelet, min		.25	3.2	Gage
106	.769 Diameter body rear of rotating band		.25	3.2	Gage
107	.10 Height of boss in tracer cavity		.25	3.2	Gage
108	.62 Depth metering disc to bottom of tracer cavity		.25	3.2	Gage

NOTES: 1/ Measurement of knurling for compliance with the minimum depth and width requirements shall apply where visual inspection indicates that the depth or width dimension is borderline.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70794(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	Body, Projectile				12551782 NEXT HIGHER ASSEMBLY 12551781
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
109	.485 Large diameter body cavity		.25	3.2	Gage
110	.010 Runout large body cavity		.25	3.2	Gage
111	1.69 Depth body cavity		.25	3.2	Gage
112	.423 Small diameter body cavity		.25	3.2	Gage
113	.005 Runout small diameter body cavity		.25	3.2	Gage
114	.087 Length boss (tracer cavity to basic diameter (igniter cavity))		.25	3.2	Gage
115	.035 Thickness igniter boss base		.25	3.2	Gage
116	.435 Diameter tracer cavity		.25	3.2	Gage
117	.008 Runout tracer cavity		.25	3.2	Gage
118	.515 Diameter metering disc cavity		.25	3.2	Gage
119	.12 Depth metering disc cavity		.25	3.2	Gage
120	.008 Runout metering disc cavity		.25	3.2	Gage
121	.517 Diameter serration in mouth		.25	3.2	Gage
122	.004 True position of serration in mouth		.25	3.2	Gage
123	.563 Diameter undercut in mouth		.25	3.2	Gage
124	.456 Depth mouth to undercut		.25	3.2	Gage
Minor					
201	.680 Diameter rotating band seat knurl		.40	3.2	Gage
202	.105 Width rotating band seat		.40	3.2	Gage
203	.082 Width knurling, minimum		.40	3.2	Visual/Gage
204	.645 Diameter mouth outside		.40	3.2	Gage
205	2.570 Overall length		.40	3.2	Gage
206	AA250 surface roughness		.40	3.2	Visual
207	Poor workmanship		.65	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70794(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551781
4.4.2.3	Projectile Body Assembly				NEXT HIGHER ASSEMBLY 12551780
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Special					
001	Ferrous metal discontinuity		100%	3.5	11075203 (ref)/ AAIE 4.4.6.1/ SPC 4.4.4
Major					
101	1.365 Length mouth to rear edge of rotating band		.25	3.2	Gage
102	.006 Runout rotating band		.25	3.2	Gage
103	.826 Diameter of rotating band		.25	3.2	Gage
104	.105 Width rotating band		.25	3.2	Gage
105	Void at band/body interface		.25	3.2	Visual
106	.19 Location crimp groove		.25	3.2	Gage
107	.074 Radius, profile crimp groove		.25	3.2	Gage
108	.005 Max radius, rear edge of crimp groove		.25	3.2	Gage
Minor					
201	Poor workmanship		.65	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70794(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Projectile Body Assembly, Painted				12551780
					NEXT HIGHER ASSEMBLY: Not Applicable
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	.786 Diameter bourrelet, max		.25	3.2	Gage
Minor					
201	Exterior paint not dry		.40	3.2	Taction
202	Marking incorrect, incomplete, illegible or missing		.40	3.2	Visual
203	Paint on rotating band incomplete or missing		.40	3.2	Visual
204	Interior paint coverage incomplete		.40	3.2	Visual
205	Poor workmanship		.65	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70794(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Disc, Closure Nozzle				12551771
					NEXT HIGHER ASSEMBLY: Not Applicable
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	.114 Through diameter		.25	3.2	Gage
102	.002 Runout .409 small outside diameter to datums A and B		.25	3.2	Gage
103	.422 Datum diameter		.25	3.2	Gage
104	.270 Length datum B to face		.25	3.2	Gage
105	.04 Length datum B to angle		.25	3.2	Gage
106	AA63 surface roughness of small diameter and angle surface		.25	3.2	Visual
Minor					
201	.010 Runout .24 counterbore, diameter to datum		.40	3.2	Gage
202	.002 Runout, .516 diameter to datums A and B		.40	3.2	Gage
203	Foreign matter		.40	3.8.2	Visual
204	Poor workmanship		.65	3.8	Visual

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Nose, Cap		12551773
			NEXT HIGHER ASSEMBLY: Not Applicable

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	.025 Nose tip thickness		.25	3.2	Gage
102	.439 Length		.25	3.2	Gage
103	1.220 Length, contour		.25	3.2	Gage
104	Outside contour		.25	3.2	Gage
105	.047 Spherical radius		.25	3.2	Gage
106	.222 Location first inside diameter from datum A		.25	3.2	Gage
107	.333 Location inside diameter from inside nose tip		.25	3.2	Gage
108	.412 diameter, large inside cylindrical		.25	3.2	Gage
109	.368 diamter, small inside cylindrical		.25	3.2	Gage
110	.528 diameter, large outside cylindrical		.25	3.2	Gage
111	.513 diameter, small outside cylindrical				
Minor					
201	Foreign matter		.40	3.8.2	Visual
202	Poor workmanship		.65	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Seal				12551778
					NEXT HIGHER ASSEMBLY: Not Applicable
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	.514 Diameter		.25	3.2	Gage
102	.0060 Thickness		.25	3.2	Gage
Minor					
201	Foreign matter		.40	3.8.2	Visual
202	Poor workmanship		.65	3.8	Visual

NOTES:

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MIL-P-70794(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Disc, Metering	1 OF 1	12551779
			NEXT HIGHER ASSEMBLY: Not applicable

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	.514 Outside diameter		.25	3.2	Gage
102	.245 Inside diameter		.25	3.2	Gage
103	.068 Thickness		.25	3.2	Gage
Minor					
201	Foreign matter		.40	3.8.2	Visual
202	Poor workmanship		.65	3.8	Visual

NOTES:

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4.4.3 Tests. The tests listed in Table II shall be conducted on each lot in compliance with the methods of inspection prescribed in 4.5. The sample shall be selected in such an manner as to be representative of the entire lot. The sample size and acceptance criteria for each test shall be as specified.

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TABLE II. Projectile tests

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70794(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Projectile, 20mm, MPT-SD, M940				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Metal Parts Security (projectile metal parts)	250	1/	3.7	4.5.1
	Band fill (body and rotating band assy, uncoated)	20	2/	3.4	4.5.2
	Phosphate Coating (projectile metal parts before painting or standard panel)	3	3/	3.6.1	4.5.3
	Paint Adhesion (projectile body assembly, painted)	13	4/	3.6.2	4.5.4
	Corrosion Resistance (projectile body assembly, painted)	32	5/	3.6.3	4.5.5
	Hardness				
	Nose, cap	80	6/	3.2	4.5.6.1
	Body, projectile	10	6/	3.2	4.5.6.2

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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Footnotes:

1/ The lot shall be rejected, if any one of the following types of metal parts separation or breakup occurs in the metal parts security test, as witnessed by hole(s) in the fragmentation screen(s), or recovery of the part(s) or fragment(s).

a. Body separation or breakup.

b. Loss of nose.

c. Complete rotating band separation. This is defined as detachment of all the band having band seat knurling imprint.

d. Partial rotating band separation. This is defined as the detachment of part of the band having band seat knurling imprint.

e. The following criteria shall apply to thrown band fragments having no evidence of band seat knurling imprint.

(1) The lot shall be rejected if three or more assemblies throw band fragments having lengths not greater than 0.500 inch.

(2) The lot shall be rejected if any assembly throws a band fragment having lengths greater than 0.500 inch (indicative of deep-seated band lamination).

(3) The lot shall not be penalized for band particles thrown as a result of normal band fringing or slivering.

2/ If two or more units of the sample fail to comply with the 50% fill requirement, the lot shall be rejected. If one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If, in the accumulated samples, two or more units fail to comply with the 50% fill requirement, the lot shall be rejected.

3/ Failure of the phosphate coating on the sample projectiles or standard panels to comply with the applicable requirements shall cause rejection of all production projectile metal parts phosphated since the preceding test.

4/ If two or more units of the sample fail to comply with the requirement for paint adhesion, the lot shall be rejected. If one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the requirement for paint adhesion, the lot shall be rejected.

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5/ If two or more units of the sample fail to comply with the requirement for corrosion resistance, the lot shall be rejected. If one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the requirement for corrosion resistance, the lot shall be rejected.

6/ If two or more units of the sample fail to comply with the hardness requirement, the lot shall be rejected. If one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the hardness requirement, the lot shall be rejected.

4.4.4 Statistical process control (SPC). Acceptance shall be in accordance with the government approved SPC plan for all special defects prescribed herein. The SPC plan shall be prepared by the contractor in accordance with the appendix of this specification. The SPC plan shall include as a minimum:

a. A control chart for the percentage of defective units (P-Chart) resulting from eddy current inspection.

b. Control charts that comply with American National Standard Institute (ANSI) Z1.1, Z1.2 and Z1.3.

c. Use of Shewhart's Eight Tests for Special Causes to determine an out-of-control process.

d. Action to be taken when the process capability (CP) and process capability index (CPK) is less than the value 1.33.

e. Sectioning of all subgroup samples rejected from eddy current AAIE.

The contractor shall submit for approval a SPC plan in accordance with the terms of the contract. See section 6.9 herein.

4.4.5 Packaging, packing and marking. Inspection for packaging, packing and marking shall comply with MIL-STD-1190.

4.4.6 Inspection equipment. The inspection equipment required to perform the tests and examination prescribed herein is described in the Paragraph Reference/Inspection Method Column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

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4.4.6.1 Automated acceptance inspection equipment (AAIE). AAIE shall be used to perform all special defect examinations and tests prescribed herein. AAIE shall be in accordance with MIL-A-70625 and approved by the government.

4.5 Methods of inspection.

4.5.1 Metal parts security. The metal parts security shall be in accordance with AMCR 715-505, Volume 8. The test cartridges, at ambient temperature, shall be fired in bursts of 50 shots in an M39 or M61 gun. The gun barrel shall be at ambient temperature for the firing of each burst. The cyclic rate will be recorded for informational purposes.

4.5.2 Band fill. After sectioning along the longitudinal axis, one half of the sectioned projectile body shall be examined for rotating band fill in the undercut areas. Failure of either of the exposed sections to comply with the 50% fill requirement shall be considered a defect.

4.5.3 Phosphate coating. The method of test shall be as prescribed in TT-C-490 for the type of coating used (see Dwg. 12551780). The prescribed sample projectiles shall be processed at the start of daily production and at 4-hour intervals, maximum, thereafter. Appearance and weight of the coating shall be recorded.

4.5.4 Paint adhesion. The method of test shall be as prescribed in TT-C-490. The test shall be made on the exterior painted surface of each projectile metal part of the sample forward of the rotating band. Visible exposure of underlying phosphate coating or ferrous metal on any unit of the sample, as a result of the test, shall be recorded as a defect.

4.5.5 Corrosion resistance. The method of test shall be as prescribed in American Society for Testing and Materials (ASTM), ASTM-B117. The paint shall not be disturbed, however, and unpainted surfaces of the sample projectile metal part may be masked out by a wax coating or other suitable material. The salt spray solution and exposure time shall be as specified in 3.6.3.

4.5.6 Hardness. The method of test shall be in accordance with ASTM-E18, Rockwell superficial hardness of metallic materials.

4.5.6.1 Nose cap. The hardness of each sample nose shall be measured in areas shown on Dwg 12551773. The hardness shall be measured using a superficial hardness tester (Amos Model 1-ST or

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equal) with direct reading on the Rockwell 15-T scale. The average of four readings shall determine hardness. No individual reading shall exceed the acceptable range (35-55) by no more than 5 points R-15T.

4.5.6.2 Body projectile hardness. The body hardness shall be measured at a minimum of seven places equally spaced along the entire side wall length to include the annealed section of the boattail on a sample of 10 sectioned units from each heat treat lot.

5. PACKAGING

5.1 Preservation and packaging. The projectiles shall be preserved and packaged in accordance with MIL-STD-1190.

5.2 Packing. The projectiles shall be preserved and packaged as specified in 5.1. Each lot shall be packed for shipment in accordance with MIL-STD-1190, as directed by the procuring activity.

5.3 Marking. Marking or labeling of packed cartons shall comply with applicable requirements of MIL-STD-1190.

6. NOTES

6.1 Intended use. The component covered by this specification is intended for use with Cartridge, 20mm, MPT-SD, M940.

6.2 Acquisition requirements. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to: Commander, U.S. AMCCOM ATTN: AMSMC-QAF-I (D), Picatinny Arsenal, NJ, 07806-5000. This address will be specified on the Contract Data Requirements Lists, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which may alter the metallurgical or physical properties of the item.

6.5 First article sample. Where a first article sample is required by the contract, the contracting officer will notify the test facility when to proceed with the metal parts security test. Additional assemblies may be required by the test facility. (see 4.3).

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6.6 Intermediate point inspection. The classification of defects identifies the defect characteristic (among other things) for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which would be detrimental to the attainment of optimum product quality of the end item.

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, U.S. Army Armament Research and Development Command (ARRADCOM) or U.S. Army Armament Reasearch and Development (ARDC) drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.8 Firing tests. In order to minimize inspection costs, the firing tests of 4.4 will be performed after the sample has been provisionally accepted by the Government for all other requirements. Additional cartridges may be required by the test facility (see 4.3).

6.9 Submission of SPC plan for approval. Submit SPC plan as required to Commander, AMCCOM, ATTN: AMSMC-QAF-A (D), Picatinny Arsenal, NJ 07806-5000.

6.10 Subject term (key words) listing.

Ammunition
Bursting Charges
Explosives
Incendiary Ammunition
M940
MP-T-SD
Multipurpose Ammunition
Nose Fuzes
Projectiles
Self-Destruct Ammunition
Small Arms Ammunition
Trace Ammunition
20 Millimeter

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1305-AD14)

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APPENDIX

PREPARATION OF STATISTICAL PROCESS CONTROL PLANS

10. SCOPE

10.1. Scope. This document is intended to serve as a guide to the preparation of statistical process control (SPC) plans by government contractors. It identifies the minimum amount of information deemed necessary to adequately describe an SPC program. But the contractor is encouraged to include whatever additional information he feels will further clarify his particular situation. A specific format for the SPC plan is proposed, mainly for the purpose of standardization. The contractor is free to vary this format, provided all information called for is included and clearly identified.

10.1.2 Application. It is the intent to require an SPC plan from a contractor under the following circumstances:

a. When called for in a contract through an "E-clause".

b. When called for in the Quality Assurance Provisions of a Military Specification.

c. Whenever a situation requires intensive control of quality.

10.1.3 Organization. This document is organized as follows:

SECTION I: Preparation of SPC Plan by Contractor

PART A: SPC Management Plan
PART B: Detailed SPC Plan

20. APPLICABLE DOCUMENTS

20.1 Government documents

20.1.1 Specifications, standards, and handbooks.

MIL-Q-9858 - Quality Program Requirements

30.0 REQUIREMENTS

This section is not applicable to this appendix.

40.0 SECTION I: PREPARATION

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APPENDIX

40.1 General. The Government's intent to institute SPC in order to improve the quality of the materiel it procures will only be successful through a joint cooperative effort between the contractor and those government personnel responsible for monitoring the SPC program in question. This effort is substantially aided by formal documentation of the contractor's general SPC policies and his procedures specific to the item being procured. Such a document, to be called the SPC Plan, serves to insure that the interests of both the contractor and the government are protected. It may reveal where additional guidance by the government Product Quality Manager (PQM) may be needed. The SPC Plan, therefore, is a major source document in an SPC program and must be carefully prepared and reviewed.

40.2 Relation to MIL-Q-9858 and quality program plan. Nothing in this document is to be taken as superceding the provisions of MIL-Q-9858, which remains as the primary document governing contractor quality programs. Similarly, SPC Plans are to be considered an integral part of the contractor's overall Quality Program Plan (QPP). The purpose of calling for a separate document for the SPC portions of the quality program is to facilitate the evaluation of the provisions therein and to highlight the use of statistics in controlling production. Information from the overall QPP may in fact be repeated in the SPC Plan, but only in that the information relates specifically to SPC.

40.3 Format. This document calls for an SPC Plan in two parts. These two parts are the SPC Management Plan and the Detailed SPC Plan. Each part of the SPC Plan is described separately in the paragraphs that follow. In each case, though, specific information will be asked for in a specific order. This information should be regarded as not exhaustive, in the sense that it is the minimum amount of information deemed necessary to adequately describe that portion of the SPC Plan. In some cases, the information asked for will not be applicable. In such cases, the contractor shall indicate why that information does not apply. Furthermore, the contractor is encouraged to include whatever additional information is felt will further explain the policies and procedures.

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APPENDIX

50. PART A: SPC MANAGEMENT PLAN

The intent of the SPC Management Plan is to provide an overview of the contractor's SPC involvement, without reference to any specific materiel item or manufacturing process. It should be a relatively brief, stand-alone document and if necessary may repeat information in the general QPP.

The contractor shall include in the SPC Management Plan a paragraph dealing with each of the following topics, including information sufficient to allow an evaluator to assess the Plan without resort to any other sources:

50.1 Policy/scope.

a. Applicability. The contractor shall discuss the policies for applying SPC, including:

- What are contractor's goals and commitments regarding SPC?
- What criteria are used to decide when SPC shall be applied?
- What alternatives to SPC have been used which have successfully reduced/prevented the production of defects?

b. Applicable documents. The contractor shall indicate the prime source documents that form the basis of the SPC Program, including ANSI/ASQC standards, text books, government documents, internal documents, etc.

50.2 SPC management structure.

a. Internal organization. The contractor shall discuss the nature of the SPC structure within the corporation, including the relationship of quality to manufacturing and to the overall organizational structure.

b. SPC responsibilities. The contractor shall discuss the responsibilities of specific positions within the SPC structure. (Individuals are not to be named, only the positions.) The discussion shall be of the nature of who does what when. Typical responsibilities to be addressed shall include:

- Who performs inspections?

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- Who has authority for acting on problems?
- Who decides on corrective action?
- Who implements the corrective action?
- Who performs audits?

50.3 SPC training. The contractor shall discuss the nature and extent of the training program for SPC. Included should be the general policy for who receives training, how much, and where. Indicate whether training is at a university, trade school, or in-house. If certain disciplines receive different types of training (such as management vs. manufacturing), this should be made clear. Discuss any formal certification or qualification programs that exist, either for individuals or for the company as a whole.

If no formal training program currently exists, discuss any plans to implement one in the future.

50.4 Vendor, subcontractor/purchase controls. The requirement to implement a formal SPC Program extends beyond those manufacturing processes over which the contractor has direct control. This requirement must be passed to all subcontractors and vendors with whom the contractor deals in the manufacture of this item. In this regard, the contractor shall discuss the policy for ensuring the use of SPC by the suppliers or subcontractors. How consistent are the vendor's procedures with the contractor's? How is it determined that suppliers have adequate controls and what degree of influence does the contractor have in modifying those controls? What provisions exist for the contractor to audit the procedures and records of subcontractors and vendors?

50.5 Manufacturing controls. SPC involves tracking performance characteristics of the product for the purpose of assessing the stability of the process. An out-of-control condition requires positive feedback to some facet of the manufacturing process in order to make the necessary adjustments. A definite portion of those adjustments relate to the manufacturing equipment itself, as opposed to raw material, personnel, or some other category. With regard to this relationship between SPC and the manufacturing equipment, the contractor shall discuss his procedures for implementing the required adjustments. Attention should be paid

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to frequency of setup, calibration, and verification. If automatic monitoring of process parameteres is used for feedback/control, is such a procedure used in lieu of SPC? Are engineering studies performed to eliminate or minimize the influence of process variation on product quality?

50.6 General SPC procedures.

50.6.1 Criteria for use of SPC. Experience has shown that SPC can effectively be applied under specific circumstances. The contractor shall discuss the criteria used to evaluate whether a particular product would benefit from SPC. Include any relationships to volume of production, classification of defects (critical/special/major/minor), degree of automation, etc. Whenever traditional SPC is not applied, what alternative procedures are used to assure quality?

50.6.2 Process capability studies. Normally, control charts are not implemented on a process until a formal capability study has been performed. The contractor shall discuss the general procedures for conducting process capability studies, especially the following topics:

- When are they used and when not?
- What is the duration of the study, in terms of sampling occasions?
- What is the timing of the study, in relation to scheduled startup of production?
- What general criteria are used for determining capability?
- What definitions of good, marginal and poor capability are generally used?
- If a process ever proves to be other than highly capable, what actions are taken?
- What steps are used to determine the true nature of the statistical distribution of the process?

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50.6.3 Control chart policy. With respect to the design and utilization of SPC control charts, the contractor shall discuss the general policies and procedures, including among others the following topics:

- What types of charts are generally used and what rationale is used for their selection?
- How are control limits established? How are they adjusted? How often are they adjusted?
- What combination of criteria are used to trigger an out-of-control signal? What criteria are used to determine non-conforming product?
- What criteria are used for the selection of sample size and frequency of sampling occasions? How are these related to production rate?
- How are control charts maintained in the instance of redundant operations, i.e. where output from identical machine or stations is potentially mixed?
- What is the policy for establishing rational subgroups?
- Has an SPC corrective Actions/Failure Analysis Program been established? Are any of its details different from those already discussed under paragraph 50.2.b (SPC Responsibilities) above?
- What provisions are in place for recording on control charts all pertinent ancillary information, such as changes in raw material supply, etc.?

50.6.4 SPC computer hardware/software. The contractor shall detail the extent to which computerized data processing and analysis ae used in his SPC effort.

- Which steps of the control process are computerized?
- Are the computer programs constant and fixed, or are they customized for each application?

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- Are computers used to make the actual decisions, or only to aid in the production of charts and reports?

- Are computer services centralized, or does each application have access to its own facilities?

50.7 Test/measurement equipment control. Controlling a process through repeated and frequent measurement of product requires a stable and accurate test and measurement system. The contractor shall explain any provisions for determining and maintaining the reliability of test and measurement equipment, when used in an SPC environment. Are different calibration and control procedures used than for equipment not involved in SPC? Is calibration information recorded on the pertinent control charts?

50.8 SPC auditing and review procedures. A major element of an SPC Program is the systematic audit and review of all critical decision points. The contractor shall discuss his provisions for performing regular audits and reviews, especially in the following areas:

50.8.1 Audit of inspection accuracy. How often and in what manner is the accuracy of inspection equipment verified? How is intervening material disposed of whenever it is found that inspection equipment has drifted?

50.8.2 Audit of control charting procedures. What provisions exist for determining the accuracy and adequacy of the control charts themselves and any subsequent decisions made based on these charts? If any discrepancies are found, how are they disposed of?

50.8.3 Reviews. How often are reviews of SPC procedures held? Who participates? Is management involved? What topics are discussed? How are proposed changes implemented?

50.9 SPC records. Thorough and efficient record-keeping is a vital facet of a responsive SPC Program. The contractor shall indicate the nature and extent of his SPC record-keeping effort, emphasizing the control and management aspects, such as who is in charge, how long are records kept, how often are they reviewed, etc. The discussion should include at least the following types of records:

a. Incoming Inspection

b. Manufacturing Inspection

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c. Vendor/Subcontractor Inspection

d. Failure Reports (internal and external)

e. Corrective Action Reports

f. Control Charts

g. Scrap and Rework Reports

h. Lessons Learned, Recommendations and Feedback

60. PART B: DETAILED SPC PLAN

The contents of the Detailed SPC Plan shall describe the specific actions to be taken in order to provide SPC on all characteristics enumerated by the government. Aside from paragraph 60.1 (Generic Procedures), which is to be included only once, the intent and design of Part B is such that each paragraph is to be included as many times as is necessary to completely describe the Plan. In other words, paragraph 60.2 (Specifics) is to be included for each characteristic called for. Each such paragraph shall include a paragraph 60.2.2 (Process Identification) for each related process to be controlled. These paragraphs in turn shall include repeated paragraphs 60.2.2.(c) (Charts to be Maintained) for each individual control chart to be maintained for that process. The Detailed SPC Plan, then, can be viewed as a tree structure, beginning with a branch for each characteristic and ending with a leaf for each control chart. Table I illustrates the recommended structure for this part of the SPC Plan.

60.1 Generic procedures. The purpose of this general paragraph is to present only once all common information which would otherwise be repeated several times in embedded paragraphs. Typical information the contractor is to place here are the make and model of any computer system used throughout the line, any major software packages, any inspection equipment used at more than one location, etc. The contractor shall also make note of any unusual circumstances, such as if portions of the line are geographically disjointed from one another, if selected personnel are to work at more than one location, if portions of the line are used to manufacture other items other than under control of this contract, etc.

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60.2 Specifics. This paragraph, along with its related subparagraphs, is to be repeated for each characteristic cited for implementation of SPC. Each such repetition shall be clearly marked with the characteristic name to properly identify it.

60.2.1 Manufacturing system for characteristic X. The contractor shall provide a brief narrative description of all pertinent details of how this characteristic is manufactured. The description shall be sufficient to allow a reviewer unfamiliar with the item to properly assess the applicability of the control measures being proposed. Included shall be such information as chain of events, sequencing in relation to other characteristics, type and number of machines involved, location of manufacturing facility if separate from the rest of the line, tolerances maintained, etc. As a minimum, the following topics shall be addressed:

a. Production Machinery

(1) How many redundant tools/machines for this characteristic?

(2) How many redundant stations on each machine?

(3) Are machines fully automatic? Semi?

(4) What is designed production rate? Will machine run continuously or in batch mode?

(5) How many shifts per day? Length of shifts?

(6) How many operators are involved in relation to the number of machines?

(7) Is the source of each in-process part or subcomponent identifiable?

(8) Is this equipment dedicated to this one characteristic, or is it shared with other lines?

(9) What is the proposed frequency of tooling changes/adjustments/calibration?

b. Inspection System

(1) How many redundant machines?

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(2) How many redundant stations on each machine?

(3) Is inspection fully automatic, semi-automatic, or manual? Likewise for gages.

(4) What is the designed inspection rate?

(5) How timely is the inspection in relation to manufacture?

(6) Is the inspection equipment operated on shifts comparable to the manufacturing equipment?

(7) Are the same personnel involved?

(8) Is the inspection system dedicated to this characteristic, or does it also do other tasks?

c. Is this characteristic produced in-house or is it contracted out? If produced by a subcontractor, who is it?

d. Capability study. Prior to implementation of SPC on a characteristic, a capability study must be conducted, the intent of which is to determine the relationship of the natural manufacturing tolerance to the specified drawing tolerance. The contractor shall discuss his proposed capability study for this characteristic, including at least the following topics:

(1) When is the study to be performed, in relation to award of contract?

(2) What sampling strategy is to be employed, in terms of occasions and sample sizes?

(3) How is the shape of the distribution determined?

(4) How is the standard deviation of the individuals calculated?

(5) What is to be done if normality does not apply?

(6) What criteria will be used to characterize capability?

(7) What will be done if the process proves to be not capable or not in control?

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60.2.2 Process identification. Repeat this paragraph for each specific process to be controlled for the above characteristic.

a. Process name and rationale. The contractor shall identify the process to be controlled and present the rationale for its selection. Attention shall be paid to such elements as narrow tolerances, lack of redundancy, history of prior difficulty, etc.

b. Capability study. The contractor shall present the details of his proposed capability study activities for this process. Note that, in general, this is a different study than the one cited in paragraph 60.2.1.(d) above, although the same guidelines apply. The capability study on the characteristic itself will normally involve reference to firm specification tolerances, whereas the capability study on the process will usually involve informal in-process dimension tolerances.

Also to be included is any prior experience with this process that might aid in evaluating the control provisons proposed.

(c) Charts to be maintained. This paragraph is to be repeated for each distinct type of chart to be used for this process. The discussion shall address at least the following points:

(1) Type of chart, to include rationale for its selection, in terms of the particular situation intended to be detected.

(2) Number to be maintained, as a function of the number of machines involved, their interdependence and equivalence, etc.

(3) Sampling procedures, to include sample size, frequency, equipment involved, basis for determination of rational subgroups, etc.

(4) Establishment of control limits, to include size of data base used, techniques employed, frequency of review, etc.

(5) Out-of-control criteria to be applied, to include the particular combination proposed and any special priority assigned to any individual criteria.

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(6) Out-of-control action to be taken, to include relation to stopping production, recalling product, replacing tooling, etc. What are the responsibilities of the individual inspectors and line supervisors?

(7) Actions to be taken when non-conforming product is found, including what actions are taken at different limits to preclude continual production of questionable quality material.

TABLE I: Example of Structure of Detailed SPC Plan

The information presented here is to serve as a clarification of the intended structure of PART B: DETAILED SPC PLAN. It illustrates the recommended manner of imbedding each required subparagraph in its proper location.

XM1 WIDGET SPC PLAN
PART B: DETAILED SPC PLAN

B.1 Generic Procedures

B.2 Characteristic A

B.2.a Description of Characteristic A
B.2.b Process 1
B.2.b.(3) Chart X
B.2.b.(3) Chart Y

B.2.b Process 2
B.2.b.(3) Chart Z

B.2 Characteristic B

B.2.a Description of Characteristic B
B.2.b.(3) Chart X

B.2 Characteristic C

B.2.a Description of Characteristic C
B.2.b Process 1
B.2.b.(3) Chart X

B.2.b Process 2
B.2.b.(3) Chart X
B.2.b.(3) Chart Y

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70794 (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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